Khushal Bhalia

+91-8657422399 khushalbhalia7@gmail.com linkedin.com/in/khush0704

Education

Indian Institute of Technology, Hyderabad — **July 2025 – Present**
M.Tech in Computer Science and Engineering — *Sangareddy, Hyderabad*

Ramrao Adik Institute of Technology — **July 2021 – June 2025**
*B.Tech in Computer Engineering, CGPA: **9.7/10*** — *Nerul, Navi Mumbai*

Experience

Nomura Finance Ltd. — **January 2025 – July 2025**
Global Markets Tech — *Powai, Mumbai*

- Worked proactively on building a tech tool for the **Fixed Income Structuring** group to **facilitate live trades**.
- Initiated a separate project in the **eFXO** division to improve **Hit and Quote Ratios** in **online FX trading** group.
- **Collaborated with cross-functional teams** of 6 engineers and the finance department to understand the financial aspects of FX trading and **translate business needs into a data-driven tool** that improved decision-making.
- Identified key factors causing low Hit/Quote Ratios by using **ML algorithms** like **DBScan to cluster similar trades**.
- Performed Static Data Analysis on a **time-series database** (**KDB**) to extract actionable insights and parsed **custom FIX (Financial Information Exchange) Protocol** to **retrieve hidden trade metadata** to devise strategic actions.
- Achieved Quote Ratio increase from **55% to 78%** and **Hit ratio from 2% to 2.8%** , driving **revenue growth**.
- Developed **Power BI dashboard** to display **trade failure reasons**, **recommended actions**, and **time-series data**.

Yuva Tutors — **June 2023 – November 2023**
Computer Science Tutor — *Nerul, Navi Mumbai*

- Worked as a **personal tutor**, teaching programming languages like **Java, and C** to 30+ higher secondary students.
- Simplified complex topics like **OOPs**, **DBMS**, and **Operating Systems improving their grades by 20%**.
- Demonstrated **adaptability and accountability** while tutoring, **consistently meeting tight schedules** alongside my regular coursework, and customizing lessons to suit individual student needs and maximize learning outcomes.
- **Fostered student interest** in foundations of CS, enhancing problem-solving skills through **hands-on learning**.

Projects

ClassQR – QR+ AI based attendance system | *Node.js, Express, PostgreSQL, Langchain, Power BI* — **April 2025**

- Engineered a real time QR code based system to instantly mark attendance by scanning the QR codes with **96% accuracy** in tracking, by using **IP filtering** and **geo-location monitoring** to prevent proxy attendance.
- Integrated an **AI-driven chatbot** to **analyze and retrieve** metrics in natural language without complex SQL queries.
- **Developed Power BI dashboards** to visualize attendance patterns, that help to **derive actionable insights**.

Sale Buddy – Real time multilingual chatbot | *Node.js, Express, Gemini AI, Bhashini API, Firebase* — **October 2024**

- A multilingual chatbot type **chrome extension** to provide support to customers of any e-commerce platform.
- It **improves instant customer support efficiency by 90%** using **In-Context Learning** model on top of Gemini AI.
- It uses the **Bhashini API** (developed by Gov. of India) to enable translation between **22 native Indian languages**.

Leaf Lens – CNN based plant disease detection | *TensorFlow, Keras, Python, NumPy, Pandas, Flask* — **January 2024**

- A web application which uses **Convolutional neural networks** to identify diseases of **40 different classes** of plants.
- Achieved high training **accuracy of 99.2%** by focusing on data quality through extensive **preprocessing** of dataset.
- Integrated a Flask-based server for real-time image upload and prediction, which could be connected to **IOT devices**.

Technical Skills

Languages: Java, Python, C, HTML, CSS, JavaScript
Cloud & DevOps: Git, Github, AWS and GCP basics
Technologies: SQL, NoSQL, Langchain, CUDA
Databases: PostgreSQL, Firebase, MongoDB, KDB
Libraries: Node.js, Express, Pandas, Numpy, Matplotlib
ML Frameworks: OpenCV, Scikit Learn, Tensorflow
Softwares: Figma, Microsoft Power BI, Excel
AI tools: Cursor, Windsurf, Claude Code, MCP server

Achievements

- Qualified till the final stage of **Code for Good** Hackathon conducted by **JPMorgan Chase.** Selected among the top 10 teams among 50K applicants. Contributed to **Open Source** by helping **Eklavya NGO** to promote higher education.
- Published a **Conference paper** titled, "**Savory Bites: A Customized Recipe Recommendation System**" at the **15th International Conference** on Science & Engineering – 2025, held at Prince Dr. K. Vasudevan College, Chennai
- Qualified **GATE CSE** twice during undergrad, and scored **99 percentile** in **GATE 2025** conducted by **IIT Roorkee**.
- Led a team to secure **top-3 position** at **Motive's GenAI Hackathon** by building a hybrid **structured–semantic video search system**, applying strong **analytical, problem-solving and leadership skills** to boost query accuracy.